

Orbie V5 electronics

what was done, how, and what to verify

Ayva Labs / Orbie · 01 October 2026 · Body board rev 0.2, dock / feeder / water / pogo rev 0.1

Status: five boards placed, autorouted and DRC-clean; fab files generated; **not yet reviewed by an engineer, not yet ordered.**

Licence: CERN-OHL-S-2.0 (hardware) · Generated designs, source in `hardware/v5/` of this repository.

Why this exists

The V3 prototype tips forward: its body sits on top of the wheel axle and the heavy head sits above that, so the centre of gravity is about 55 mm above the axle. V5 keeps the V3 shell and face and makes the robot an actively balanced two-wheeler, like Loona, with a real-time motion MCU in the body, encoder motors, an IMU, and the dock, battery-management and sensing items from the 26 September 2026 hardware-requirements document. This package is the electronics for that: a body controller that retrofits the V3 body, a belly pogo pad, a dock base, and two plug-in dock modules.

What we ask the reviewers

1. Confirm the body-board outline against the body you have (section 6: the March STEP does not contain the January "Main PCB").
2. Check the five component values flagged in section 7 against current datasheets.
3. Eyeball the autorouted switching loops (MT3608, DRV8833) and the two sense-resistor connections.
4. Decide 1S2P vs 2S, ballast vs lighter head, and nose-in vs reverse-in docking (section 6).

1. Where V5 came from

Left: the V3 body as built (barrel on the axle, 120 mm head, caster and front skids). Right: the same robot family re-cut as a wheeled balancer. The painted prototype of 29 March 2026 is the body this package retrofits.



V3 painted prototype, 29 Mar 2026, with the front skids added to stop it falling forward

Architecture: two brains

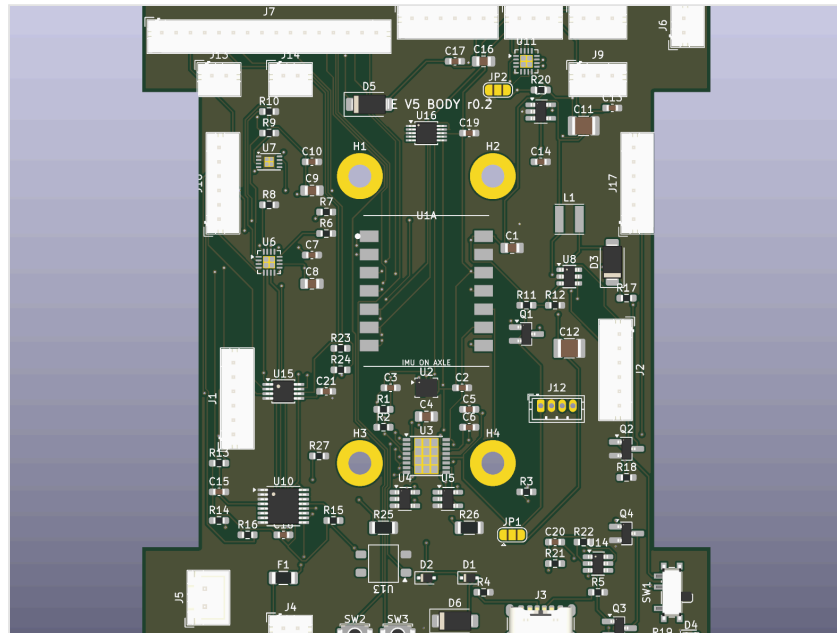
- **Head (unchanged V3 head board):** XIAO ESP32-S3 Sense. Camera, Wi-Fi/BLE, dot-matrix face, laser, servo PWM, I2S audio out, app API, OTA.
- **Body (new board):** XIAO RP2040. 1 kHz balance loop from the IMU, quadrature encoders on PIO, locked-antiphase PWM into the DRV8833, cliff and current sensing through an I2C ADC, bumper input. Motor safety timeouts live here and never on anything Linux-based (REQ-C3).
- **Link:** one UART over the neck harness. The head sends velocity setpoints; the body answers with pitch, speed, battery and faults. A future Linux SoC for the pet-health models talks to the same UART.
- **Dock:** XIAO ESP32-C3 with its own Wi-Fi. Feeding runs on schedule whether or not the robot is alive. Modules are dumb.

2. The five boards

Body controller r0.2 (orbie_v5_body)

V3 Main PCB outline 79.5 × 94.62 mm (outline to be confirmed), 2-layer, 1481 tracks, DRC 0, ERC 0

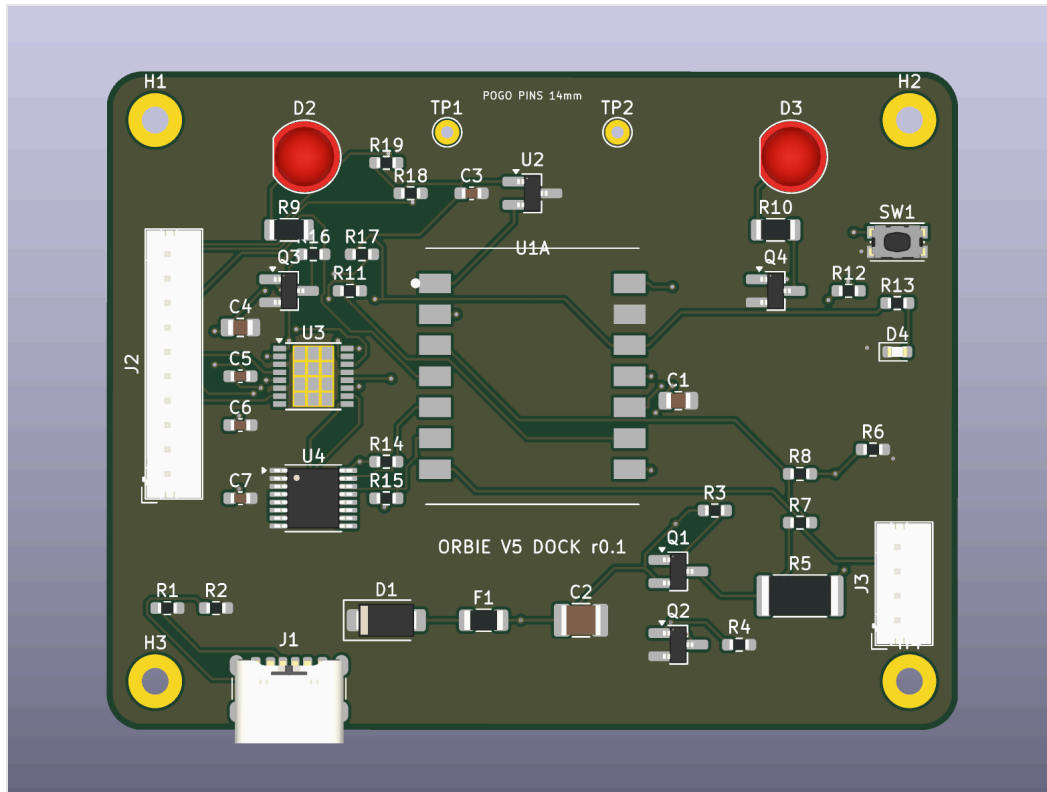
XIAO RP2040 motion MCU · LSM6DS3 IMU · ADS1015 (cliff IR ×2, motor current ×2) · DRV8833 locked-antiphase + 74LVC1G04 ×2 · GA12-N20 encoder ports ×2 · BQ24074 charger/power path · DW01A + AO3400A ×2 pack protection · NTC port · LC709203F gauge · MT3608 5 V · AP2112K 3V3 · TCA9517A I2C buffer · TCA9534 expander · VL53L5CX ports ×3 · MAX98357A + gain strap · USB-C 6P + TVS · pogo input · LD19 lidar port · servo P-FET · hall · 38 kHz IR receiver · RESET/BOOT side buttons · 16-pin neck harness



Dock base r0.1 (orbie_v5_dock)

70 × 54 mm, 2-layer, 528 tracks, DRC 0, ERC 0

XIAO ESP32-C3 · USB-C 5 V in + TVS + 3 A polyfuse · AO3401A pad switch with 100 k pull-down and 0.05 Ω shunt read by the C3 ADC (pads dead until hall + handshake) · P50 pogo pins at 14 mm · IR beacon LEDs ×2 (L/R lobes, 38 kHz) · AH1806 hall · DRV8833 for feeder motor + pump · TCA9534 · 10-pin stacking bus · feed/pair button · status LED

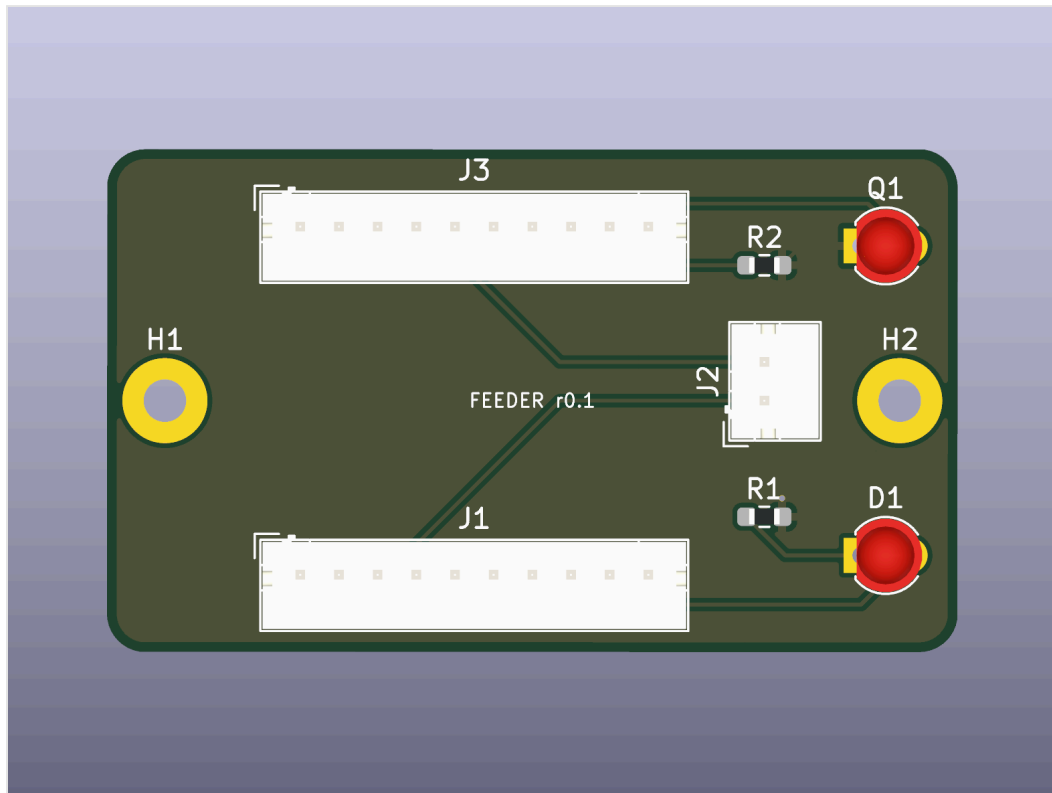


Dock base r0.1: top side, autorouted, ground pours on both layers

Feeder module r0.1 (orbie_v5_feeder)

44 × 26 mm, 2-layer, DRC 0

bus in (bottom) and bus out (top) for stacking · paddle-motor JST · 940 nm drop-beam LED + phototransistor across the chute · PRESENT_F strap. No MCU.

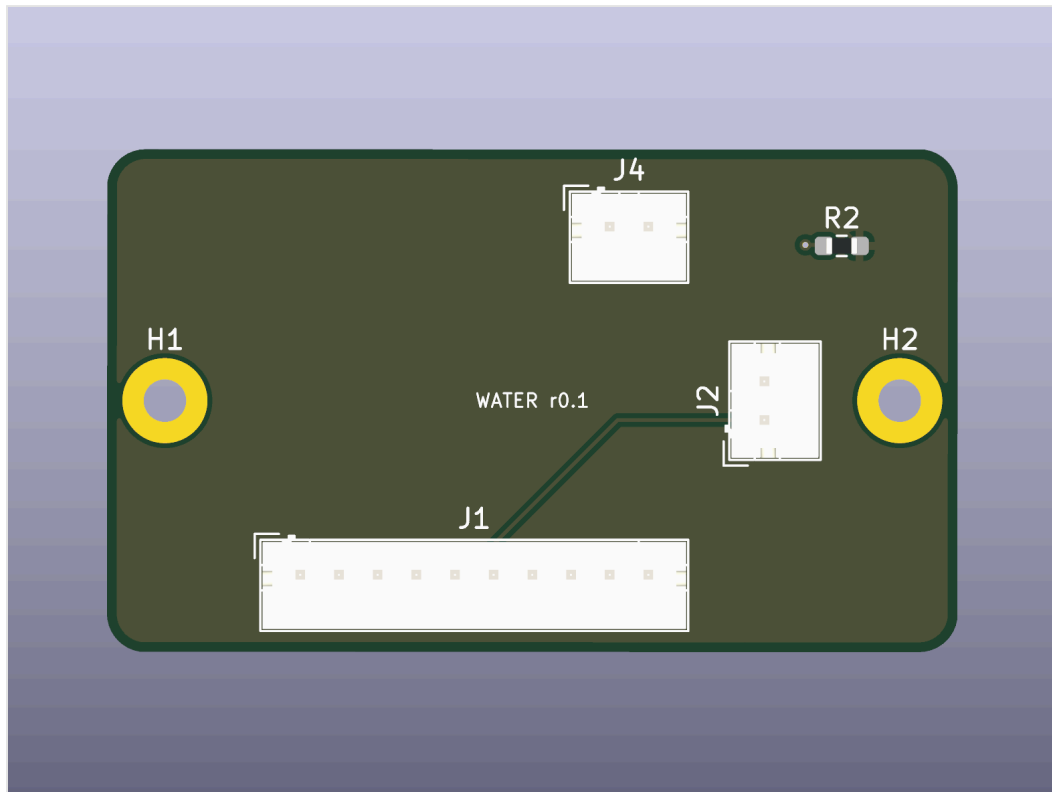


Feeder module r0.1: top side, autorouted, ground pours on both layers

Water module r0.1 (orbie_v5_water)

44 × 26 mm, 2-layer, DRC 0

bus in · pump JST · float switch (low level) · PRESENT_W strap. Nothing electrical in the water path.

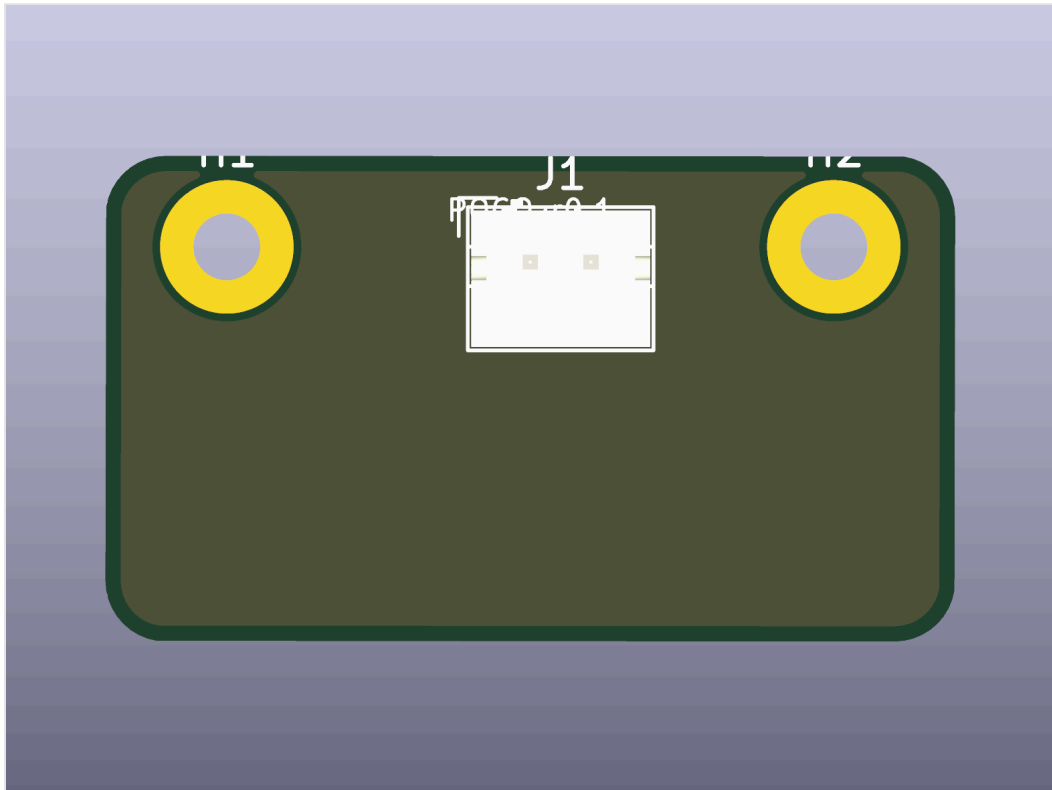


Water module r0.1: top side, autorouted, ground pours on both layers

Belly pogo pad board r0.1 (orbie_v5_pogo)

28 × 16 mm, 2-layer, DRC 0

two Ø6 mm gold pads at 14 mm, facing down through a window in the body floor; JST to the body J4



Belly pogo pad board r0.1: top side, autorouted, ground pours on both layers

3. Requirement coverage, retrofit and build notes

1 Oct 2026, rev 0.2 of the body board, rev 0.1 of dock / feeder / water boards. Generator: `kicad/gen_v5.py`.

How the boards are designed (the process, honestly)

1. **Part choice** comes from my working knowledge of the datasheets of common parts (TI BQ2407x, DRV8833, TCA9534/9517, Maxim MAX98357A, ST LSM6DS3 / VL53L5CX, onsemi LC709203F, Fortune DW01A, Sseed XIAO pinouts). I do not fetch datasheets live while generating; values such as the BQ24074 ILIM/ISSET/ITERM resistors, the MT3608 feedback divider and the DRV8833 sense resistor are computed from the datasheet formulas I know. **Each value must be checked against the current datasheet before the order** — that is on the review list below.
2. **Symbols and footprints** come from the stock KiCad 10 libraries, which encode pin names, pin numbers and package dimensions from the manufacturers' drawings. A Python model (`gen_v5.py`) assigns every pin a net by name, so a wrong pin name fails loudly at generation time instead of silently in layout.
3. **Placement** is by hand in the generator (coordinates in mm), checked by KiCad DRC until courtyards, holes and edges are clean.
4. **Routing** is freerouting (autorouter) from a Specctra export, imported back, then GND pours on both layers, then KiCad DRC again. The router works to JLCPCB's 2-layer limits (0.127 mm track/space, 0.3 mm drill).
5. **Fab outputs** are produced by `kicad-cli`: gerbers incl. paste layers (stencil), drill, pick-and-place, and a BOM. LCSC part numbers are *not* filled in; the MPN column is.

What that process is good at: correct netlists, correct packages, no shorts, consistent boards. What it is not: a layout engineer's judgement on the switching loops (MT3608, DRV8833) and the sense-resistor Kelvin connections. Those four spots want five minutes from Sumit before gerbers go out.

Requirement coverage

Req	What it asks	Where it is	Status
2.1 / REQ-C1..C3	RAM; SLAM off the ESP32; defined MCU↔SoC link, motor timeouts on the MCU	Body board: RP2040 owns motors with its own watchdog; UART to the head; a later Linux SoC talks to the same UART	design rule satisfied; firmware
2.2 / REQ-N3	VL53L0X unreliable, replace	3 × VL53L5CX connectors (front, left, right) with LPn lines from the expander for address assignment	done (option A)
2.3 / REQ-V1	camera	head board, not this revision; OV5640 swap recommended	open
2.4 / 2.5 / REQ-A1..A4	speaker distortion, gain, sample rate, HPF	GAIN strap JP2 (GND 12 dB / open 9 dB / VDD 6 dB / R20 3 dB); amp on the clean 5 V rail; 48 kHz is a firmware change; speaker spec "sealed back 4 Ω 3 W" in BOM	done (hardware part)
2.6 / REQ-R3	thermal	not solved by a board; the Linux SoC decision drives it	open
2.7 / REQ-X1..X6	reset / boot without disassembly	SW2 RESET and SW3 BOOT, side-actuated, behind rear pinholes; XIAO RST/BOOT carried on harness pins 14/15; 10 s hold = factory reset in firmware; status LED	done (hardware part)
2.8 / REQ-R5	USB brownout / back-feed	D5 Schottky between body 5 V and the head so the XIAO's USB cannot source into a laptop; TVS on VBUS; CC pull-downs	done
REQ-S1	topple → motor cut < 200 ms	IMU on the RP2040 bus; RP2040 drives PWM to 50 % (brake) and can drop nSLEEP via expander	firmware on done hardware
REQ-S2	750 mm drop	mechanical	open

Req	What it asks	Where it is	Status
REQ-S3	self-recovery / announce	firmware + speaker	open
REQ-S4	cliff sensors	J10/J11 downward IR, analog into ADS1015 AIN0/1	done
REQ-S5	contact detection	bumper J13 on RP2040 D10 and motor current on 0.1 Ω sense into ADS1015 AIN2/3	done (both)
REQ-N1	encoders ≥ 20 cpr	GA12-N20 encoder motors, 6-pin J1/J2, PIO decode	done
REQ-N2	IMU to navigation	heading in the UART telemetry	firmware
REQ-N4	lidar in the base	J12 LD19 port: 5 V, TX $\rightarrow$ head GPIO4 via harness pin 13	done (port only; lidar not in BOM)
REQ-N5 / T1 / T5	dock beacon + contacts, docked = hall AND charge current	IR receiver U13 rear; hall U12; charger CHG/PGOOD on expander	done
REQ-R1	I2C integrity	TCA9517A buffer between body bus and the neck harness, 2k2 pull-ups each side, modules on JST	done
REQ-R4	60 min navigation	1S2P 18650 (V3 pack); budget to be measured	open
REQ-P1	pack protection independent of charger	DW01A + 2 $\times$ AO3400A on the pack negative	done
REQ-P2	charge temperature interlock	J14 NTC to BQ24074 TS; R9 bypass marked DNP	done
REQ-P3	matched cells	procurement note	open
REQ-P4 / P5	shutdown and reserve policy	LC709203F gives battery_pct ; policy is firmware	firmware
REQ-D1	dock pads dead until robot present	Dock: AO3401A high-side switch, 100 k pull-down, 0.05 Ω shunt read by the C3 ADC, hall + handshake before enable	done
REQ-D2	dock must not slide	mechanical (mass / rubber foot)	open
REQ-D3	feeder works without the robot	dock has its own XIAO ESP32-C3 with Wi-Fi and schedule; feeder is a dumb module	done
REQ-D4 / D5	food safety, spill path	mechanical; motors are in the modules behind the bus, nothing electrical in the water path	partly, mechanical open
REQ-D6	jam / empty / over-dispense	drop-beam sensor on the feeder board; limits in dock firmware	hardware done
REQ-T2..T4	approach clearance, retry, blocked alcove	firmware + manual	open

The five boards

Board	Size	What	State
orbie_v5_body r0.2	V3 Main PCB outline (confirm)	motion MCU, IMU, ADC, charger, protection, gauge, 5 V, 3V3, expander, I2C buffer, amp, 3 ToF, lidar, servo, hall, IR, reset/boot, 16-pin harness	routed, DRC 0
orbie_v5_pogo	28 $\times$ 16	belly pads	routed
orbie_v5_dock	70 $\times$ 54	XIAO ESP32-C3, safe pads, IR beacon $\times 2$, hall, DRV8833 for feeder + pump, 10-pin stacking bus, button, LED	routed
orbie_v5_feeder	44 $\times$ 26	bus in/out, paddle motor, drop beam, PRESENT_F	routed
orbie_v5_water	44 $\times$ 26	bus in, pump, float switch, PRESENT_W	routed

Stacking bus (JST PH 10-pin on every module, spring contacts on the decks): 5V, GND, FEED_A, FEED_B, PUMP_A, PUMP_B, DROP_SENSE, PRESENT_F, LEVEL_SENSE, PRESENT_W.

Neck harness J7 (16-pin)

1 GND · 2 +5V_HEAD · 3 +3V3_E · 4 SDA_H · 5 SCL_H · 6 UART_RP_RX (XIAO GPIO1) · 7 UART_RP_TX (GPIO2) · 8 I2S_BCLK (7) · 9 I2S_LRC (8) · 10 I2S_DIN (44) · 11 SERVO_PWM (43) · 12 IR_RX (GPIO3) · 13 LIDAR_RX (GPIO4) · 14 XIAO_RST · 15 XIAO_BOOT · 16 GND

Review list before ordering

1. **Outline.** The 8 Mar 2026 STEP has no 79.5×94.62 board and no 18.6×40 bosses; the body holds the 73×43 ESP-ROLL board. Ajay to confirm what the body you have mounts, then I re-outline in ten minutes.
2. BQ24074: ILIM 1k1 / ISET 590 / ITERM 1k5 / TMR 10k, and TS behaviour with a 10 k NTC — confirm against the current datasheet rev.
3. DW01A thresholds vs the chosen cells; AO3400A as the protection FETs (5.7 A, fine for a 1S2P pack at 1.5 A charge / ~3 A peak).
4. ADS1015 reading 0–200 mV across the 0.1Ω sense: use the ± 256 mV range.
5. MT3608 feedback 75 k / 22 k $\rightarrow$ 5.0 V; inductor $4.7 \mu\text{H} \geq 3$ A sat.
6. Sumit to eyeball the autorouted loops around U3, U8 and the two sense resistors.

4. Retrofit into the V3 body, harness and order list

30 Sep 2026. STATUS: body board routed (2-layer, 986 tracks), DRC 0 errors, ERC 0 errors; pogo board routed. Gerbers incl. F.Paste (stencil), drill, pick-and-place and BOM in `kicad/v5/out/` . Files: `kicad/v5/` . Generator: `kicad/gen_v5.py` , router/fab script: `kicad/route_v5.py` .

What this board is

The V3 body "Main PCB" replaced 1:1 by a balancing controller. Same outline (79.5 × 94.62 mm, 63.74 mm waist), same four M3 holes at 18.6 × 40 mm, so it screws onto the existing posts in the `Body_Bottom1` shell. The V3 head board with the XIAO ESP32-S3 Sense, display and sensors is **not** changed; only the neck harness pinout changes (see below).

Two-brain split:

Board	MCU	Does
Head (V3, unchanged)	XIAO ESP32-S3 Sense	camera, Wi-Fi/BLE, face, laser, servo PWM, I2S audio out, app API, OTA
Body (this board)	XIAO RP2040, soldered flat on castellations	1 kHz balance loop: LSM6DS3 IMU on its own I2C, two quadrature encoders on PIO, two locked-antiphase PWM channels into a DRV8833, UART to the head

The head sends velocity setpoints (`drive` , `turn`) over UART; the body replies with pitch, speed, battery and fault flags. If the head reboots for an OTA the robot keeps standing.

Retrofit into the V3 body: what fits, what to reprint

Item	Fits the existing body?	Action
PCB	yes, same outline and holes	none
Motors	no , plain N20 has no encoder	GA12-N20 6 V with magnetic encoder, 300–500 rpm. Same 12 mm gearbox, fits the <code>Motor_C</code> clamp ; the encoder disc adds ~9 mm at the back of the motor toward the body centre. Check against the battery support; if it touches, move the pack up 10 mm (it wants to be high anyway).
Caster ball + front skids	must go	leave the caster socket empty, remove the skids
Rear rest foot	new	small TPU foot on the rear lower shell so it can kneel with motors off. Glue-on for the rig.
Pogo pads	new	28 × 16 mm pad board under the floor; needs two Ø7 mm windows 18 mm apart in the bottom shell. Drill/file for the rig; new print of <code>Body_Bottom1</code> later.
USB-C	rear edge of the board	slot in the rear shell at board height, or leave for bench charging with the shell off
Speaker	front slot as V3	JST PH2 on the front band
ToF	front + both sides	3 × VL53L5CX modules on 6-pin JST (J15/J16/J17); cliff IR modules on J10/J11 at the front edge facing down
Battery	1S2P 18650 (the March CAD already holds two cells)	XH2 connector through DW01A protection; NTC on the cells to J14; mount it high (above the axle) for a balancer
Harness	new pinout	16-pin JST PH, below

So: **no new body print for the rig**. The only printed part that changes for the product is the bottom shell (pogo windows + foot).

Neck harness (J7, JST PH 16-pin, rev 0.2)

Pin	Net	Head side (XIAO ESP32-S3)
1	GND	GND
2	+5V_HEAD	5V pin, through D5 so the XIAO USB can never back-feed a laptop
3	+3V3_E	head sensors if they want the body rail
4	SDA_H	GPIO5 (through the TCA9517A buffer)
5	SCL_H	GPIO6
6	UART_RP_RX	GPIO1 (TX) — was motor A_FWD
7	UART_RP_TX	GPIO2 (RX) — was motor A_REV
8	I2S_BCLK	GPIO7
9	I2S_LRC	GPIO8
10	I2S_DIN	GPIO44
11	SERVO_PWM	GPIO43
12	IR_RX	GPIO3 — was motor B_REV
13	LIDAR_RX	GPIO4 — was motor B_FWD
14	XIAO_RST	XIAO reset pad (SW2 on the body, rear pinhole)
15	XIAO_BOOT	XIAO boot pad (SW3; hold 10 s = factory reset in firmware)
16	GND	GND

GPIO9 stays the laser.

Motor drive: locked antiphase

`M0T_L_PWM` goes to DRV8833 AIN1 and, through a 74LVC1G04 inverter, to AIN2. 50 % duty = stop, above = forward, below = reverse. One RP2040 pin per motor and no dead-zone at zero speed, which is what a balance loop wants. Cost: some current ripple at zero speed, so run PWM at 20 kHz and expect the motors to be slightly warm when idle. `JP1` picks VM = VSYS (battery, ~3.7 V) or +5 V boost for more torque on carpet.

Firmware to write (not in this repo yet)

- RP2040 (`orbie_motion`): PIO quadrature x2, LSM6DS3 at 1 kHz, complementary filter, cascaded PID

(angle inner, speed outer), UART protocol, fall detect, kneel/stand state machine.

- ESP32-S3 `d_wip` changes: remove LEDC motor code; add UART1 on GPIO1/2 to the body; `/motor` becomes a

setpoint forward; add `/api/motion` telemetry; RMT IR receiver on GPIO3.

Order list for the rig (2 boards)

Qty	Part	Note
2	XIAO RP2040	Seeed 102010428
4	GA12-N20 encoder motor 6 V 300–500 rpm	with 6-pin cable
2	LSM6DS3TR-C	LGA-14
2	DRV8833PWPR	HTSSOP-16
2	BQ24074RGTR	VQFN-16
2	LC709203FQH-01TWG	WDFN-8
2	TCA9534PWR	TSSOP-16
2	MAX98357AETE+T	QFN-16
2	MT3608, AP2112K-3.3, AH1806-W, IRM-H638T	SOT / SC-59 / THT
4	74LVC1G04 SOT-23-5	
2	AO3400A, AO3401A	SOT-23
2	SS14, 4 PMEG2005EJ	
2	NR4018 4.7 μ H inductor	
2	GCT USB4125 6-pin USB-C	
2	JST PH B12B, B6B $\times$ 2, B3B, B2B $\times$ 3, XH B2B	
2	PCM12 slide switch, 1206 2 A polyfuse	
—	0603 R/C kit, 0805 10 μ F, 1210 22 μ F $\times$ 4	
2	ADS1015IDGS, TCA9517ADGKR, DW01A-G, 4 $\times$ AO3400A extra, SMAJ5.0A $\times$ 2	rev 0.2 additions
6	VL53L5CX breakout (Pololu / ST SATEL)	3 per robot
4	downward IR reflective sensor module, analog out	cliff
2	10 k NTC bead, 2 $\times$ 0.1 Ω 1206 1 W, 2 $\times$ Alps SKRK side tact	rev 0.2
2	JST PH B16B (neck), B6B $\times$ 3 (ToF), B3B $\times$ 2 (cliff), ZH B4B (lidar)	rev 0.2 connectors
1	Dock set: XIAO ESP32-C3, 2 $\times$ 940 nm 5 mm IR LED, 2 $\times$ MMBT2222A, 0.05 Ω 2512, P50 pogo pins $\times$ 2, DRV8833, TCA9534, PH B10B $\times$ 3, PH B2B/B4B	dock + modules
2	JLCPCB stencil, top side, 0.12 mm	order with the gerbers (body + dock on one panel if possible)

Full list: [kicad/v5/out/orbie_v5_body_bom_jlc.csv](#) (add LCSC numbers there if you want JLC to assemble instead).

Build order in the office

1. Stencil paste, place all SMD on the top side, hot plate at 165 °C (low-temp paste) or 230 °C.
2. Bottom side: hall sensor and the two 0603 by hand.
3. Through-hole: JSTs, USB-C, switch, IR receiver, pin header.
4. Solder the XIAO RP2040 last, flat on its castellations.
5. Power on with no motors: check VSYS, +5V, +3V3_E, +3V3_RP, then I2C scan from the head (expander 0x38, gauge 0x0B) and from the RP2040 (IMU 0x6A).
6. Motors on the bench, then encoders, then the balance loop with the shell open and a hand ready.

5. Verification performed

Check	Tool	Result
Electrical rules (every pin assigned or marked no-connect; power pins driven)	KiCad 10 ERC, all severities	body 0, dock 0, feeder 0, water 0, pogo 0 errors
Design rules after routing (clearance 0.15 mm, track 0.2 mm, via 0.6/0.3, courtyards, edge clearance, mask bridges, starved thermals)	KiCad 10 DRC, all severities	0 errors on all five boards
Unrouted connections	KiCad DRC	0 on all five boards
Outline and hole pattern	read 1:1 from the V3 "Main PCB" drawing DXF	matched; but see section 6
Mechanical keep-outs from the drawing (7.5 mm over the waist, 1 mm on the bottom motor-clamp zones)	manual	respected
3D fit of encoder motors, harness entry, pogo window	—	not done
Datasheet check of computed values	—	not done
Firmware for RP2040 / dock C3 / head changes	—	not written

6. The outline question (blocker for the body board)

The body board copies the outline and the four M3 holes from the drawing *Main PCB* (24 Jan 2026): 79.5 × 94.62 mm, 63.74 mm waist, holes at 18.6 × 40 mm. To verify it I loaded the assembly STEP of 8 March 2026 (Pawme_Assembly.STEP, 639 solids) in FreeCAD and looked for that board and for screw bosses on that pattern.

- There is **no** 79.5 × 94.62 board in the March assembly. The body contains the ESP-ROLL board (Rolling+Robot+Top+Board, 73.0 × 42.7 mm) and the display board (80 × 36 mm).
- No boss pattern in Body_Top, BodyTop_Subassembly or Support_Subassembly matches 18.6 × 40 mm. Patterns found: 62 × 44.8 mm (r 1.1) on the battery support, 47.4 × 46.8 mm (r 1.4) and 75 × 80 mm (r 1.32) on the top shell.
- The March assembly also shows **two** 18650 cells side by side (so 1S2P is what the body already holds), a 42 × 22 × 52 speaker, and Ø89.8 mm wheels on a 129 mm track.

So either the body you have was built to the March CAD (then the board must be re-outlined to its mounts), or the January Main PCB posts exist in a later shell revision. Please tell us which, or send the current Body_Bottom STEP. The outline is one function in the generator; re-cutting it is a ten-minute change and re-routing is automatic.

7. Review checklist for the engineers

1. **Outline / mounting** (section 6). Blocking for the body board only.
2. **BQ24074**: ILIM = 1k1 (1.5 A), ISET = 590 Ω (1.5 A), ITERM = 1k5, TMR = 10 k, and TS with a 10 k NTC (0–45 °C window; R9 bypass marked DNP). Confirm against the current datasheet revision and the cells chosen.
3. **DW01A** thresholds (2.4 V / 4.3 V / 150 mV OC) against the cells; AO3400A as the protection switches (5.7 A).
4. **ADS1015** input range for the 0.1 Ω motor sense (use the ± 256 mV PGA) and for the cliff IR analog outputs (0–3.3 V, ± 4.096 V range).
5. **MT3608**: 75 k / 22 k feedback $\rightarrow$ 5.0 V; 4.7 μ H $\geq$ 3 A inductor; 22 μ F in/out.
6. **Locked-antiphase drive**: 20 kHz PWM, expect idle current ripple; confirm the N20 encoder motors accept it, or switch firmware to sign-magnitude on the same hardware.
7. **Switching loops**: U8/L1/D3/C11 and U3/C4 were placed tight but autorouted; check loop area and that AISEN/BISEN sense returns are Kelvin to the shunt pads.
8. **Neck harness**: 16 wires through the neck; confirm the V3 head board can expose XIAO RST/BOOT (pins 14/15) and GPIO4 for the lidar.
9. **Dock pads**: AO3401A + shunt + firmware fold-back is the safety path for REQ-D1; decide whether a dedicated eFuse (TPS2553 class) is wanted for production.
10. **Thermal**: nothing on these boards addresses the 58 °C shell reading; the companion-SoC decision drives it.

8. How these boards were designed

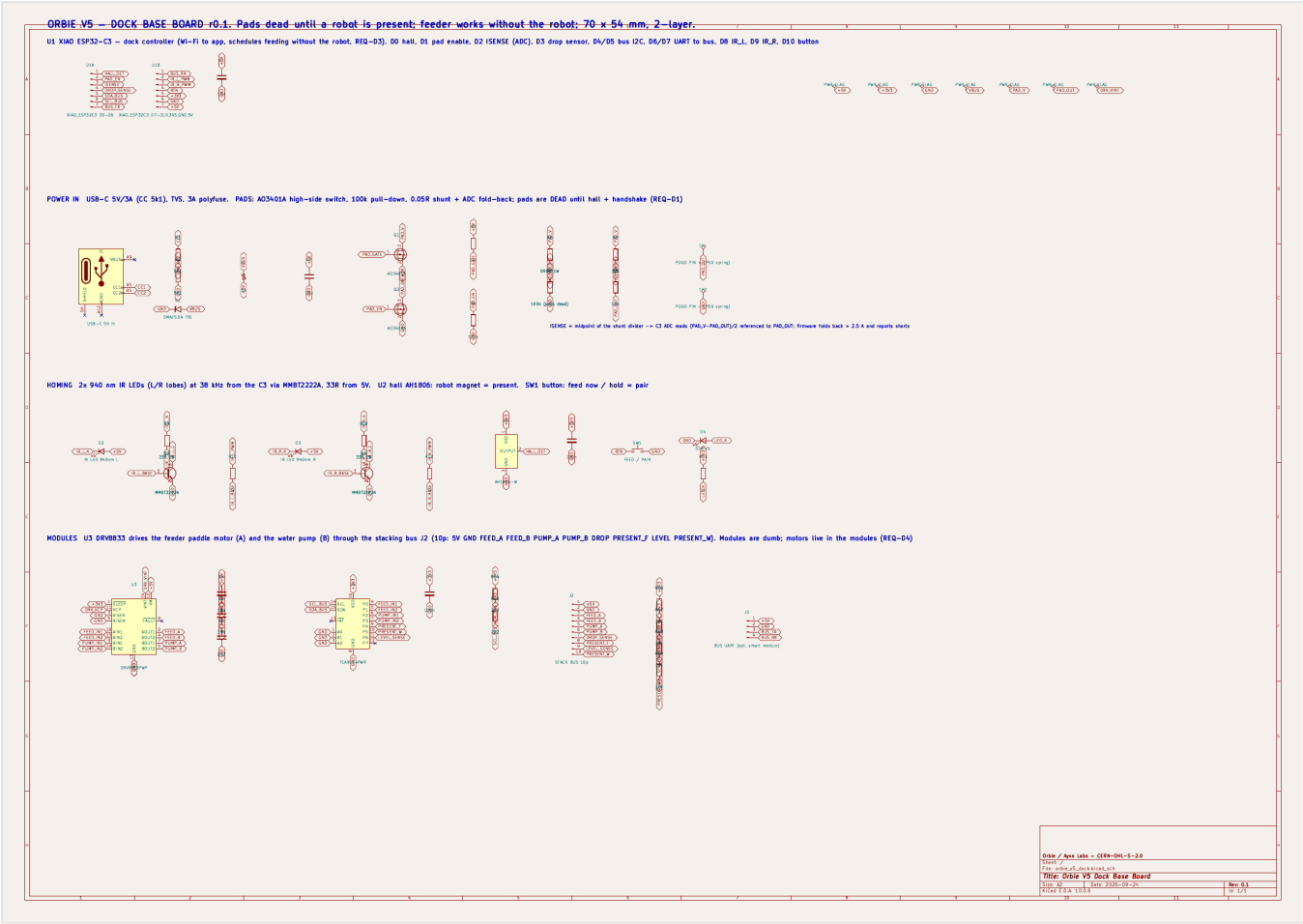
- **Parts** were chosen from working knowledge of the common parts' datasheets (TI BQ2407x, DRV8833, TCA9534/9517; Maxim MAX98357A; ST LSM6DS3, VL53L5CX; onsemi LC709203F; Fortune DW01A; Seeed XIAO pinouts). Datasheets were not fetched live; computed values are listed above for checking.
- **Symbols and footprints** are the stock KiCad 10 libraries (pin names, pin numbers, package dimensions from the manufacturers' drawings). A Python model (`gen_v5.py`) assigns every pin a net by name, so a wrong pin name fails at generation time.
- **Placement** was done by hand in millimetres in the generator and iterated against DRC until courtyards, holes and edges were clean.
- **Routing**: freerouting (autorouter) via Specctra DSN/SES, then GND pours on both layers, then DRC again, to JLCPCB 2-layer limits (0.127 mm track/space, 0.3 mm drill).
- **Outputs**: `kicad-cli` gerbers including paste layers (stencil), Excellon drill, pick-and-place CSV, JLC-format BOM (LCSC column empty, MPN filled).
- Everything is regenerable: edit the generator, re-run, re-route. Do not hand-edit the outputs unless you also stop using the generator.

Full-size PDFs are in `hardware/v5/out/*.sch.pdf`; these pages are previews.

[illegible]

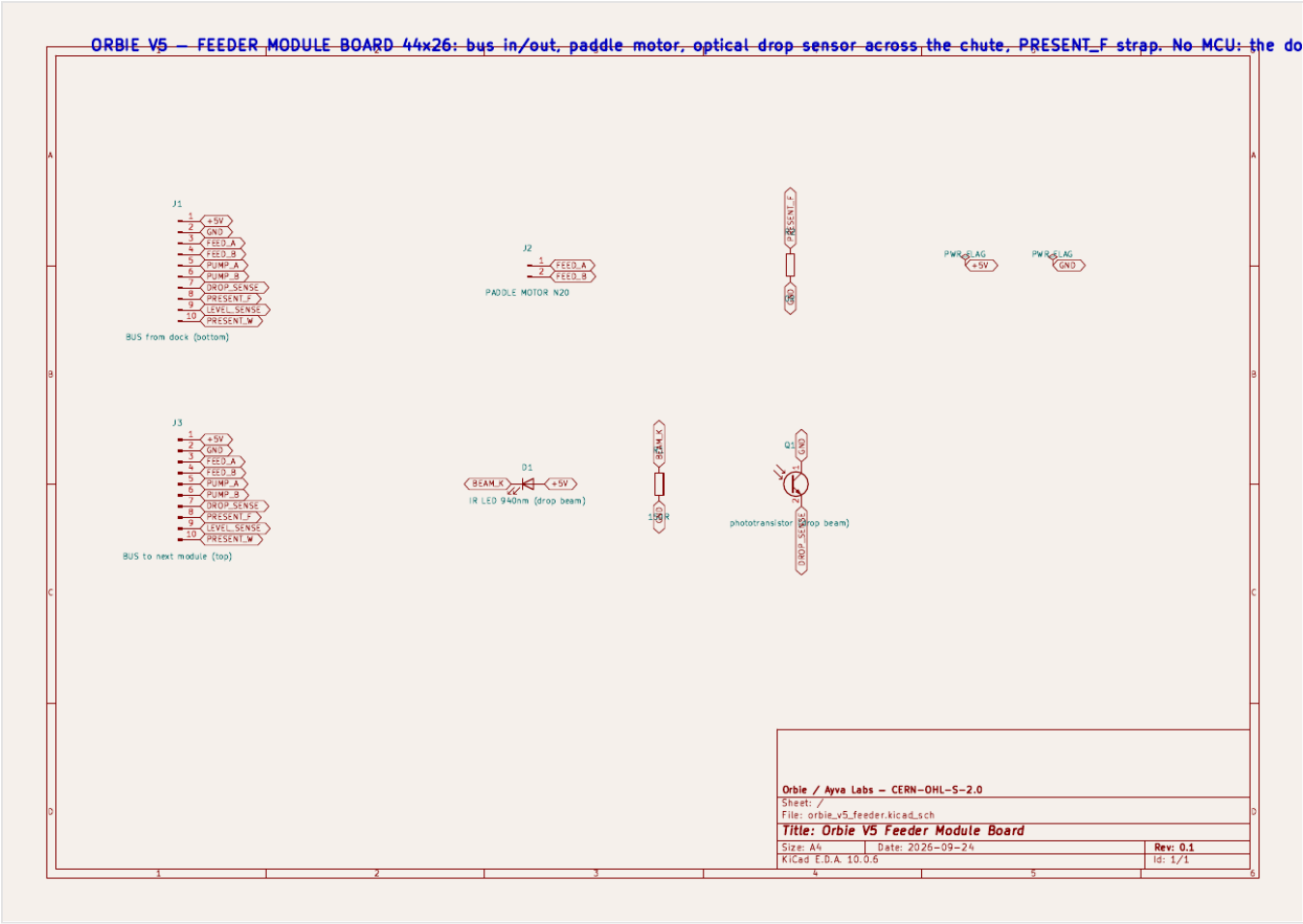
orbie_v5_body.kicad_sch

Dock base r0.1



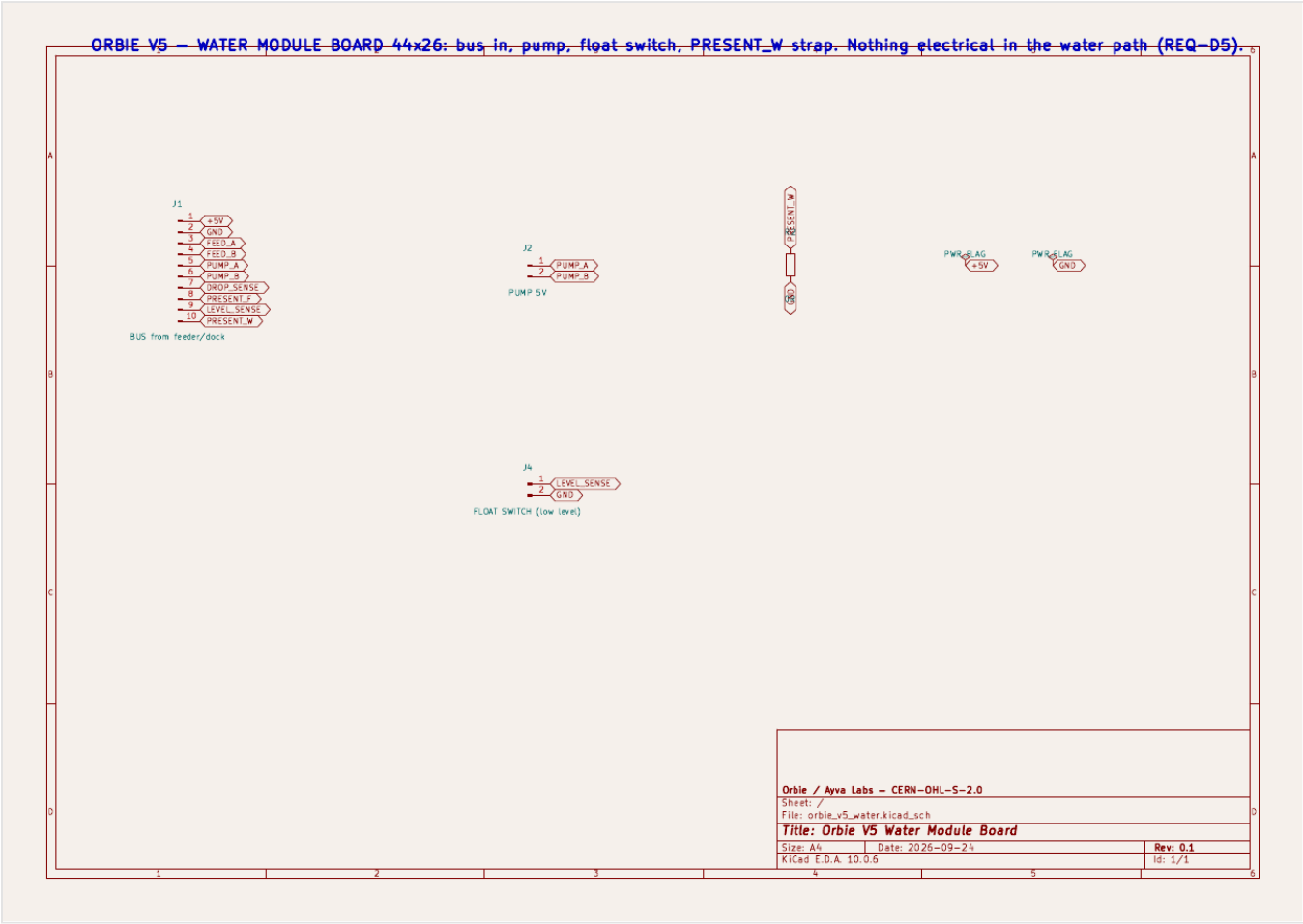
orbie_v5_dock.kicad_sch

Feeder module r0.1



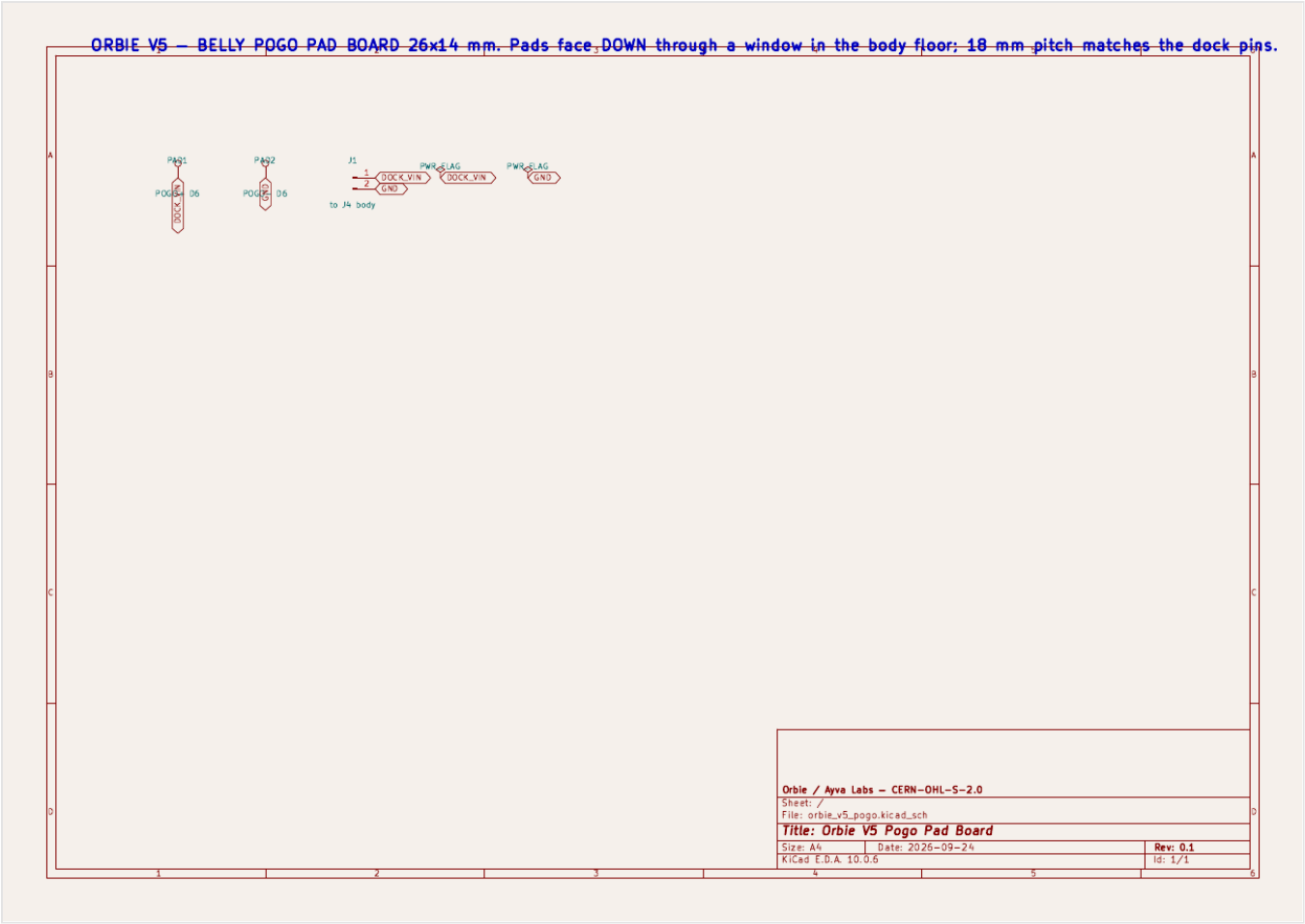
orbie_v5_feeder.kicad_sch

Water module r0.1



orbie_v5_water.kicad_sch

Belly pogo pad board r0.1



orbie_v5_pogo.kicad_sch

Appendix B. File index

Path (repo: hardware/v5/)	What
gen_v5.py, gen_kicad.py, route_v5.py, orbie_v5.pretty/	generator, library helpers, router/fab script, custom footprints (XIAO castellated, pogo pad)
orbie_v5_<board>.kicad_pro/.kicad_sch/.kicad_pcb	KiCad 10 projects, routed
out/<board>_gerbers.zip	gerbers (F/B Cu, paste, mask, silk, edge), drill, job file
out/<board>_pos.csv, out/<board>_bom_jlc.csv	pick-and-place, BOM
out/<board>.sch.pdf, .top.png, .bottom.png	schematic PDF, 3D renders
out/<board>.erc.json, *.drc.json	check reports
docs/V5_SPEC_COVERAGE.md, docs/V5_BODY_RETROFIT.md	sources of sections 3 and 4